Lesson 6

INNOVATIONS IN FIELD EXTENSION

6.1 Introduction

Of late, a lot of emphasis is being given on innovative extension approaches at the field level, in order to ensure an effective implementation of the Transfer of Technology (ToT) process. Many such new initiatives have been undertaken in a lot of countries, the details of which are as under:

6.2 Farmer Field Schools - A New Participatory Approach

The farmer field school is a form of adult education, which evolved from the concept that farmers learn optimally from field observation & experimentation. FFS is a tool to build capacities of farmer groups through participatory approach for promoting sustainable agricultural development, managing crop eco system, to make them better decision makers in sustainable use of resource at the cropping, farming & watershed levels.

The term Farmer Fields Schools came from the Indonesian expression *Sekolah Lapangan* meaning just *field schoo*l. The first Farmer Field School was established in 1989 in Indonesia for control of Brown Plant Hopper pest in rice through Integrated Pest Management (IPM).

Although Farmer Field Schools were designed to promote IPM, empowerment has an essential feature from the beginning. The curriculum of the FFS was built on the assumption that farmers could only implement IPM once they had acquired the ability to carry out their own analysis, make their own decisions and organize their own activities. The empowerment process, rather than the adoption of specific IPM techniques, is what produces many of the developmental benefits of the FFS.

6.3 Participatory Extension Approaches

Participatory extension provides a framework for extension staff to participate with village Communities in facilitating development planning and activity implementation. This approach ensures the extension response becomes community driven and assist village communities implement their planned activities with routine monitoring and evaluation of activities and development progress. Importantly, as the name implies, the extension process is seeking maximum participation from women and men from all groups within the target village communities. People's participation implies the active involvement in development of the rural people, particularly disadvantaged groups that form the mass of the rural population and have previously been excluded from the development process. Experience has shown that through participatory programmes and activities it is possible to mobilize local knowledge and resources for self-reliant development and, in the process, reduce the cost to governments of providing

development assistance. People's participation is also recognized as an essential element in strategies for sustainable development of agriculture.

People's participation should be viewed as an active process in which people take initiatives and action that is stimulated by their own thinking and deliberation and which they can effectively Influence. Participation is therefore more than an instrument of implementing government projects. It is a development approach which recognizes the need to involve disadvantaged segments of the rural population in the design and implementation of policies concerning their well-being. While participatory approaches have been successful in many countries at stimulating self--help activities at the local level, they can and should also be followed in the design, implementation and evaluation of large scale projects.

6.3.1 Different participatory approaches

Participation aims at bringing about change in people's attitude which is critical on the part of the people involved towards their environment and adoption of interventions for agricultural development. In recent years, there have been a lot of developments, in the use of participatory approaches. Some of these approaches focus more on problem diagnosis, other are more oriented to community empowerment, some concentrate on facilitating farmer-led research and extension, while others are designed to get professionals in the field to listen to farmers. Some of the these approaches are;

6.3.1.1 Rapid rural appraisal (RRA)

RRA is a social science approach that emerged in the early 1980s for applications in development cooperation. In it, a multidisciplinary team makes use of simple, nonstandard methods and the knowledge of local people to quickly elicit, analyze and evaluate information and hypotheses about rural life and rural resources that are of relevance for taking action. RRA techniques are an attractive alternative to conventional survey methods when the aim is not to systematically capture precise figures, a typically time-consuming and cost-intensive undertaking, but rather speedy and action-oriented assessment of local knowledge, needs and potentials with an aim to elaborating strategies to resolve conflicts or investigate specific problems. They are also suitable for shifting the focus of conventional surveys onto essential aspects.

6.3.1.2 Participatory rural appraisal (PRA)

PRA is a way of enabling local (rural and urban) people to analyze their living conditions, to share the outcomes and to plan their activities. It's a handing over the stick to the insider in methods and action. The outsider's role is that of a catalyzer, a facilitator and convener of processes within a community, which is prepared to alter their situation.

6.3.1.3 Participatory learning approaches (PLA)

PLA is an approach for learning and organizing participation of local communities and groups for interacting with them, understanding them and learning from them. It helps in initiating a participatory process, in sustaining it and in opening up vistas of avenues for participation. It is a means of identifying and facilitating intended groups and evoking their participation and also opening ways in which such groups can participate in decision-making, project design, planning, execution and monitoring.

6.3.1.4 Participatory learning methods (PALM)

PALM uses the key RRA (rapid rural appraisal) concepts, but emphasizes participation by village residents and the function of the externals as catalysts and partners for self-determined development. The aim of PALM is to go beyond appraisal and arrive at participatory analysis and a common understanding of rural conditions. The focus is on learning from and with local people in order to avoid stimulating false expectations.

6.3.1.5 Agro-ecosystem analysis (AEA)

An agro ecosystem is created by the combination and inter-reaction of ecological and socioeconomic processes. Analysis of the diagrams gives rise to a table of the most important factors influencing the system properties. The system properties are: **1)** productivity (net output of an evaluated product), **2)** stability of productivity under normal conditions, **3)** sustainability of productivity under stress, and **4)** equitable distribution of production and the resources and inputs required to achieve it.

6.3.1.6 Participatory action research (PAR)

PAR is learning-by-doing approach in which the investigator places his/her knowledge at the disposal of local groups. The following is expected of an action researcher: awareness of one′s own limitations and value orientation; willingness to empathize with and share the problems and needs of local people; knowledge of their history and political and economic situation. The action researcher then engages in a dialog with the local groups, and works together with them in small discussion groups to search for solutions to the their problems

Lesson 7

EXTENSION LEARNING PROCESS

7.1 Extension Learning Process

Any change of behaviour which takes place as a result of experience may be called learning.

'Learning' is the process by which an individual, through his own activity, attains a change in his behaviour. It is an active process on the part of the learner. The essential role of an extension worker is to create effective 'learning situations'. An effective learning situation requires the following essential elements.

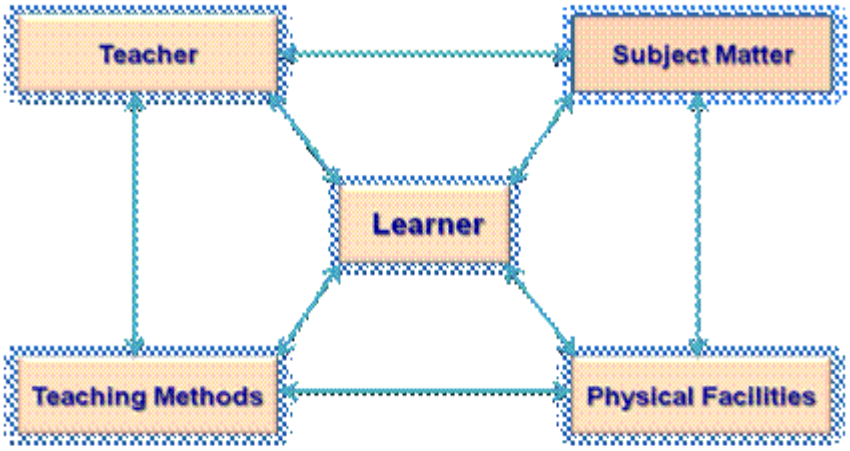


Fig. 8.1 Elements of learning situation

7.2.1 Teacher / agric. Officer/ animal husbandry officer (Instructor)

A successful teacher is the one who takes into account the following important considerations:

- ✓ understands the nature of the learner and is in a position to evaluate his motives & ability.
- ✓ Reacts appropriately to the feelings, emotions and attitudes of the learners.
- ✓ Is conscious of the fact that the learner responds to the whole gamut of learning situation.
- ✓ Is professionally alert.
- ✓ Is well-composed even in the adverse conditions.
- ✓ Has the ability to maintain better human relations.
- ✓ Always makes efforts for his professional growth.
- ✓ Selects those experiences which suit to the abilities and needs of the learners.
- ✓ Should be close to masses and belong to the same groups for making dialogues easier.

7.2.2 Learners

They should have:

- ✓ Clear objectives regarding the use of the subject matter.
- ✓ Interest in the subject matter.
- ✓ Need for the subject matter.
- ✓ Ability to understand and opportunity to apply the subject matter.
- ✓ Physical and mental preparation.

7.2.3 Subject matter (contents)

The subject matter should fulfil the following conditions:

- ✓ Should be in the line with the needs and interests of the learners.
- ✓ Should be valid, authentic, factual and applicable.
- ✓ Should be well-organized.
- ✓ Should be at a level where it could be understood by the learners.
- ✓ Should be timely & appropriate.

7.2.4 Teaching methods

Subject matter cannot effectively be transferred without the help of suitable teaching methods and aids. Proper selection and skilful handling of teaching methods and aids facilitate creation of desirable learning situation. Teaching methods & aids should be:

- ✓ Readily available.
- ✓ In working conditions.
- ✓ Suitable to the subject-matter.
- ✓ Diversified, flexible and suited to the needs of the clients.

The following points need to be kept in mind while using extension methods for appropriate involvement of different sense organs:

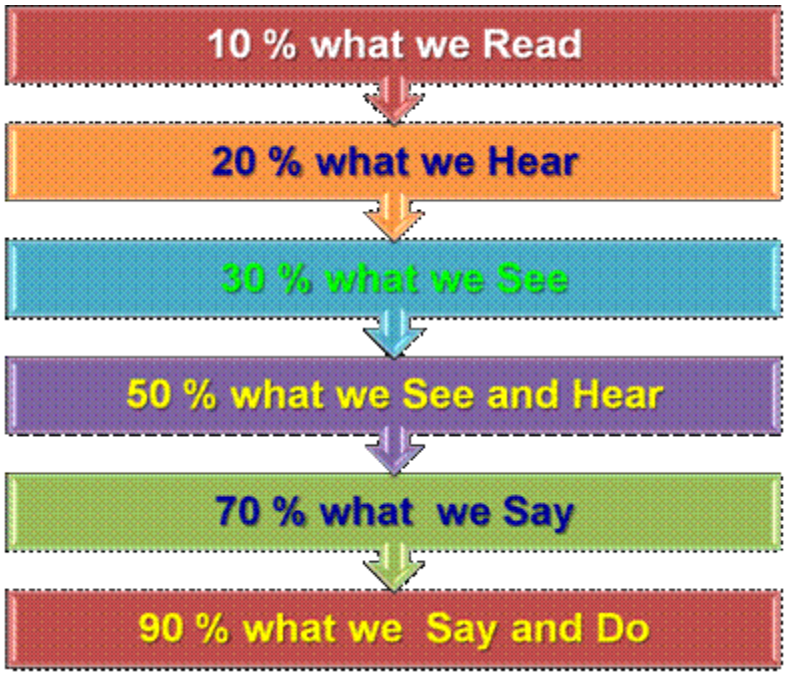


Fig. 8.2 People remember the things

7.2.5 Physical facilities

Light, ventilation, chair and table arrangements, outdoor facilities, gardening & good land escaping should be a satisfying and comfortable, if possible. The good teacher will make all these arrangements prior to the meetings.

The teacher should skilfully manipulate the elements of the learning situations & provide satisfactory learning experiences for the stakeholders. The main aim of the teacher is to bring about a change in the behaviour of the learners with the help of a judicious combination & use of different elements.

Lesson 8

TEACHING PROCESS IN EXTENSION

8.1 Introduction

'Teaching' is the process of arranging situations in which the things to be learnt are brought to the notice of the learners, their interest is developed & desire aroused, i.e. they are stimulated to act. For example, if we want to teach the dairy farmers the use & advantages of milk processing ,we do this by conducting demonstrations at the household levels, showing them how to boil the milk and at what temperature citric acid is to be added, etc. Then, comparison of the selling cost of boiled milk and raw milk is to be done to make them understand its benefits. After seeing the beneficial effect of aboiled milk, the farmer is convinced & motivated for action to start the processing of milk.

8.2 Teaching in Extension

The development and progress that we aim for is based on what we know, what we think, what we actually did, and what we can do with our physical, technological and human resources. To progress, we must not stop at the present conditions of living and must take the necessary actions to improve them on a sustainable basis. In this process, we must identify and perpetuate only the useful and successful developmental initiatives from the past, exercise concern only for the promising developmental ideas in the present, and focus on all of these practices on achieving a better and sustainable development for the future. Thereby, we gain the ability to substitute the 'good and new' developmental ideas with the 'old and outdated' through learning. Thus, learning is most effective when done under organized teaching.

Teaching is defined as an interactive process primarily involving classroom talk, which takes place between teacher and pupil, and occurs during certain definable activities.

3.3 Steps of Teaching

Teaching, in the context of extension and development, can be thought of as providing purposeful direction through learning process. Teaching *per se* is not giving knowledge or skills to people or communities, instead teaching is the process of providing opportunities to people or communities to produce relatively permanent change through their engagement in learning experiences that are provided by teachers.

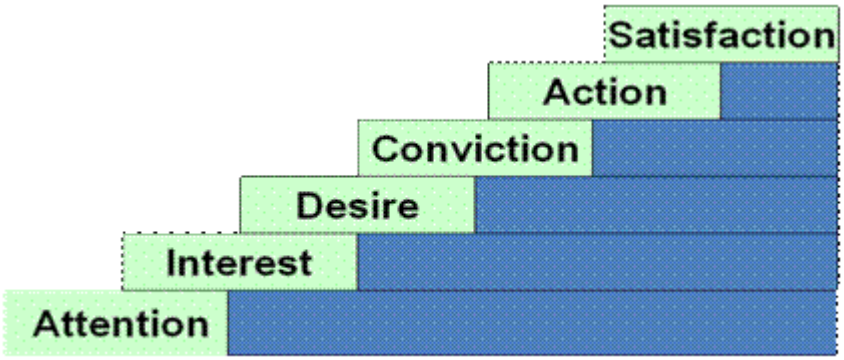


Fig. 9.1 Steps in extension teaching (Wilson and Galup, 1955)

8.3.1 Attention

Attention is the starting point to arouse the interest of the learners. Research suggests that the attention of people is attracted by various senses in the following proportions .

Table 9.1 Attention of people through various sense organs

Sense	Percent
Seeing	87.00
Hearing	7.00
Smell	3.50
Touch	1.50
Taste	1.00

The famous extension saying is, *seeing is believing*. Thus, seeing and hearing are the major senses involved to attract attention and enhanced learning.

8.3.2 Interest

Once attention is captured, instructors and/or extension professionals can bring the audience's attention to developmental needs and arouse their interest in further consideration of ideas. Extension professionals should make them understand how development contributes to the overall well-being of the community.

8.3.3 Desire

It concerns about the continuation of the audience's interest in the developmental ideas or better practices, until that interest becomes a desire, or, a motivating force.

8.3.4 Conviction

In this step, people know what action is necessary and just how to take that action. Extension professionals also ensure that people visualize the action in terms of their own situations and acquire confidence in their own ability to participate in the people-centred developmental initiatives.

8.3.5 Action

Unless this conviction is converted into action, the efforts of extension for development will go unrewarded. It is the job of extension professionals to make it easy for the people to act. For example, if the adoption of a new high yielding cattle breed is the action needed by farmers, that breed should be made available within the reach of farming communities, along with other recommended package of practices. If the action does not quickly follow desire and conviction, the new idea may fade away. Therefore, this phase should never be ignored.

8.3.6 Satisfaction

Satisfaction is the end product of extension teaching process. Follow up action(s) by extension professionals helps people to learn and evaluate the development progress. The saying, *satisfied customer is the best advertisement*, also applies to extension and development work. Satisfaction helps the people to continue developmental work with increased attention, interest, desire, conviction and action.

The six steps in teaching discussed as above, often blend with each other. As an extension professional, you need to arrange the learning situations in all the six teaching steps with the help of suitable extension teaching methods and audio-visual aids. Various teaching methods and audio-visual aids are not equally suited for every step in teaching. Every method and aid under certain circumstances makes a contribution to each step. It depends on the extension professional how he handles the situation.

Lesson 9

EXTENSION TEACHING METHODS

9.1 Introduction

Extension teaching methods are the tools & techniques used to create situations in which communication can take place between the rural people & the extension professionals. These are the methods of imparting new knowledge & skills to the rural people by drawing their attention towards such technologies, thereby arousing their interest and helping them to have a successful experience of the new practice. A proper understanding of these methods and their selection for a particular type of work is necessary.

9.2 Classification of Extension Teaching Methods

9.2.1 According to use

One way of classifying the extension methods is according to their use & nature of contact *vis-a-vis* interpersonal, group and mass communication. Based upon the nature of contact, they are classified into following categories:

9.2.1.1 Individual-contact methods

Extension methods under this category provide opportunities for face-to-face or person-to-person contact between the rural people and the extension professionals. These methods are very effective in teaching new skills and creating goodwill between farmers and the extension professionals.

9.2.1.2 Group-contact methods

Under this category, the rural people or farmers are contacted in a group which usually consists of 20 to 25 persons. These groups are usually formed around a common interest. These methods also involve a face-to-face contact with the people and provide an opportunity for the exchange of ideas, for discussions on problems and technical recommendations. In this way, the future course of action is finalized.

9.2.1.3 Mass or community-contact methods

An extension professional has to approach a large number of people for disseminating information and helping them to use it. This can be done through mass-contact methods conveniently. These methods are more useful for making people aware of the new technologies, quickly.

Table 10.1 Classification of extension-teaching methods according to their use

Individual contacts	Group contacts	Mass contacts
Farm & home visits	Method and result demonstration	Print media (viz. Newspapers, magazines, leaflets, posters, pamphlets, circular letters bulletins)
Office calls	Group meetings, discussions	Electronic media (viz. Radio, television, cell-phones)
Telephone calls	Conferences, seminars, workshops, etc.	Internet-based media
Personal letters	Field trips, field days, campaign	Exhibitions,

9.2.2 According to form

Extension teaching methods are also classified according to their forms, such as written, spoken & audio-visual. Some of the important methods under each of these 3 categories are listed in table 10.2.

Table 10.2 Classification of extension-teaching methods according to their form

Written	Spoken	Object or visual
Bulletins	General & special meetings	Result demonstrations
Leaflets, folders, news articles	Farm & home visits	Demonstration plots
Personal letters	Official calls	Motion-pictures or movies, charts
Circular letters	Telephone calls, radio	Slides & film-strips, models, exhibits

A brief description of some of the extension methods which are commonly used by extension professionals is given below:

9.2.2.1 Farm & home visit

Farm & home visit constitutes the direct or face-to-face contact by an extension professional with the farmer or the members of his family. During these visits, information is exchanged or discussed. The visits may be to get acquainted with the problems of the farmers. Such visits provide an opportunity for a two-way communication.

9.2.2.2 Method demonstration

It is used to show the technique of doing things or carrying out new practices e.g. clean milk production, paneer making, ghee making, etc. This method is usually used for groups of people.

9.2.2.3 Result demonstration

Result demonstration is meant for proving the advantages of recommended practices and to demonstrate their applicability to the local conditions. It is conducted by a farmer under the direct supervision of an extension professional. It is designed to teach others, in addition to the person who conducts the demonstration. It helps the farmers to learn by seeing & doing. This method can be used to show the superiority of practices, such as dahi culture, value addition, hygienic handling of animals, etc.

9.2.2.4 Group discussion

All the farmers cannot be contacted by extension professionals individually because of their large number. It is convenient & feasible to contact them in groups. This method is commonly known as group discussion. It is used to encourage & stimulate the people to learn more about the problems that concern the community through discussion. It is a good method of involving the local people in developing local leadership & in deciding on a plan of action in a democratic way.

9.2.2.5 Exhibition

An exhibition is a systematic display of information, actual specimens, models, posters, photographs, and charts, etc. in a logical sequence. It is organised for arousing the interest of the clientele in the things displayed. It is one of the best media for reaching a large number of people, especially illiterate & semi-literate people. Exhibitions are used for a wide range of topics, such as planning a model village, showing high-yielding breeds of cattle and buffaloes, new agricultural implements and the best products of village industries.

9.2.2.6 Campaign

Campaign is used to focus the attention of the people on a particular problem, e.g. milk adulteration, vaccination and prevention of animal diseases, tick control etc. Through this

method, maximum number of farmers can be reached in the shortest possible time. It builds up community confidence and involves the people emotionally in a programme.

9.2.2.7 Field tour

Conducted tours for farmers are used to convince them and to provide them with an opportunity of seeing the results of new practices and products, skills, etc. and to give them an idea regarding the suitability & application of these things in their own area. Such tours may also be arranged to enable the rural people to visit places & institutions (connected with the problems of rural life), such as research institutions, training institutions, agricultural universities, model villages, areas of advanced developments, leading private farms, exhibitions, and agricultural & cattle fairs/*dairy mela*.

9.2.2.8 Print media

Newspapers, magazines, bulletins, leaflets, folders, pamphlets and wall news-sheets are another set of mass media for communicating information to a large number of literate people. They are used for communicating general & specific information on a programme of technology or a practice. Small folders, leaflets & pamphlets are used to give specific recommendations about a practice, such as clean milk production, vaccination schedule, detection of milk adulterants, etc.

9.2.2.9 Radio

It is one of the most powerful media of communication. It is a mass medium of communication and can reach a large number of people at any given time involving the least expense. Extension professionals use the radio for communicating information on new methods & techniques, giving timely information about the control of animal diseases such as foot & mouth disease (FMD), animal pests, weather, market news, etc. For this purpose, talks, group discussions, folk-songs, dialogues & dramas are usually broadcast. There are radio programmes broadcast by All India Radio (AIR), FM (frequency modulated) radio, community radio, etc.

9.2.2.10 Television

It combines both audio & visual impact and is very suitable for the dissemination of agriculture & dairy information. It is more useful in teaching to do a specific job. A beginning has been made in India for using this medium for development programmes and it is expected that its use will become more extensive in the coming years. At present, along with the Government-owned channel (Doordarshan), several other private channels are telecasting various kinds of entertainment and developmental programmes to reach the viewers.

9.2.2.11 Leaflet

A leaflet is a single sheet of paper used to present information on only one developmental idea in a concise manner, using simple language.

9.2.2.12 Folder

A folder is a single piece of paper folded once or twice, and, when opened, the material is presented in sequence.

9.2.2.13 Pamphlet

A pamphlet is an unbound single sheet of paper that is printed on both sides, printed in colours with action photographs, giving full information about a topic in greater length than in folders or leaflets.

9.2.2.14 Bulletin

A bulletin is a publication of around 20 pages, with the primary objective of giving complete information which the intended readers can apply to their own situation.

9.2.2.15 Booklet

When the extension material exceeds 20 pages and is less than 50 pages, it is called booklet.